

HW 4

obs. values

1) $\text{VAR} \begin{bmatrix} z \\ M \\ y \end{bmatrix}$ $X \in \mathbb{R}^{n \times p}$ obs. of X

(full) $y = \beta_0 \mathbf{1}_n + \beta_1 z + \beta_2 M + \beta_3 X + \epsilon y$

est. "direct effect" of z on y

short
(full)

indirect effect of z on y thru m

$y = \tilde{\beta}_0 \mathbf{1}_n + \tilde{\beta}_1 z + X \tilde{\beta}_2 + \tilde{\epsilon} y$

difference estimator $\tilde{\beta}_1 - \beta_1$

(mean)

$M = \gamma_0 \mathbf{1}_n + \gamma_1 z + X \gamma_2 + \epsilon M$

product estimator $\hat{\gamma}_1 \hat{\beta}_2$

prove $\tilde{\beta}_1 - \beta_1 = \hat{\gamma}_1 \hat{\beta}_2$

(that the difference estimator + product estimator are identical)

$\hat{\beta}_1 = (X_1^T X_1)^{-1} X_1^T y$

$\tilde{\beta}_1 =$

column 1?

$\begin{pmatrix} \tilde{\beta}_0 \\ \tilde{\beta}_1 \\ \tilde{\beta}_2 \end{pmatrix} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} + \beta_3 \begin{pmatrix} d_0 \\ d_1 \\ d_2 \end{pmatrix}$

how to obtain d ?
get in form...

residuals w/ respect to x

$$(full) \quad y^* = \beta_1^{\wedge} z^* + \beta_2^{\wedge} M^*$$

$$(short) \quad y^* = \tilde{\beta}_1^{\wedge} z^*$$

$$M^* = \delta_1^{\wedge} z^* + u$$

sub into full

$$y^* = \beta_1^{\wedge} z^* + \beta_2^{\wedge} (\delta_1^{\wedge} z^* + u)$$

$$y^* = \beta_1^{\wedge} z^* + \beta_2^{\wedge} \delta_1^{\wedge} z^* + \beta_2^{\wedge} u$$

$$y^* = (\beta_1^{\wedge} + \beta_2^{\wedge} \delta_1^{\wedge}) z^* + \beta_2^{\wedge} u$$

compare

$$(short) = y^* = \tilde{\beta}_1^{\wedge} z^*$$

$$\tilde{\beta}_1^{\wedge} = \beta_1^{\wedge} + \beta_2^{\wedge} \delta_1^{\wedge}$$

$$\tilde{\beta}_1^{\wedge} - \beta_1^{\wedge} = \beta_2^{\wedge} \delta_1^{\wedge} \quad \checkmark \text{ OK}$$

2) R^2 or r_{xy} from Th. 10.1

Goal: $R^2 = \hat{\rho}_{\hat{y}\hat{y}}$ where

$$\hat{\rho}_{\hat{y}\hat{y}} = \frac{\sum (y_i - \bar{y})(\hat{y}_i - \bar{y})}{\sqrt{\sum (y_i - \bar{y})^2} \sqrt{\sum (\hat{y}_i - \bar{y})^2}}$$

$$R^2 = \frac{\sum^n (\hat{y}_i - \bar{y})^2}{\sum^n (y_i - \bar{y})^2} \quad (\text{definition})$$

$$R^2 = \left(\frac{\sum^n (y_i - \bar{y})(\hat{y}_i - \bar{y})}{\sqrt{\sum^n (y_i - \bar{y})^2} \sqrt{\sum^n (\hat{y}_i - \bar{y})^2}} \right)^2$$

(numerator) $(y_i - \bar{y})(\hat{y}_i - \bar{y})$

$$y_i = x_i \beta - \varepsilon$$

$$(\hat{y}_i - \varepsilon - \bar{y})(\hat{y}_i - \bar{y})$$

$$y_i = \hat{y}_i - \varepsilon$$

$$\hat{y}_i - \hat{y}_i \bar{y} - \varepsilon \hat{y}_i - \varepsilon \bar{y} - \bar{y} \hat{y}_i + \bar{y}^2$$



$$(\hat{y}_i - \bar{y})^2 - \cancel{\varepsilon(\hat{y}_i + \bar{y})} \quad \text{since } \varepsilon \perp \hat{y}_i$$

$$R^2 = \left(\frac{\sum (\hat{y}_i - \bar{y})^2}{\sqrt{\sum (\hat{y}_i - \bar{y})^2} \sqrt{\sum (y_i - \bar{y})^2}} \right)^2$$

↓ cont

$$= \frac{\sum (y_i^1 - \bar{y})^2}{\sum (y_i^1 - \bar{y})^2 \quad \sum (y_i - \bar{y})^2}$$

$$= \frac{\sum (y_i^1 - \bar{y})^2}{\sum (y_i - \bar{y})^2}$$

✓ OK

3) Standardized + studentized residual
p. 11.6:

$$1) (n-p-1) \hat{\sigma}_{[-i]}^2 = \frac{(n-p) \hat{\sigma}^2}{1-h_{ii}}$$

defn of
variance
estimator

$$\hat{\sigma}^2 = \frac{1}{n-p} \sum_{i=1}^n \hat{\epsilon}_i^2$$

$$\hat{\sigma}_{[-i]}^2 = \frac{1}{n-p-1} \sum_{k \neq i} \hat{\epsilon}_k^2$$

from OLS identity of LOO

$$\hat{\beta}_{[-i]} = \hat{\beta} - \frac{(X^T X)^{-1} x_i \hat{\epsilon}_i}{(1-h_{ii})}$$

$$\hat{\epsilon}_{[-i]} = y_i - x_i^T \hat{\beta}_{[-i]}$$

$$\hat{\epsilon}_{[-i]} = \frac{\hat{\epsilon}_i}{1-h_{ii}}$$

$$\hat{\epsilon}_{[-i]}^2 = \frac{\hat{\epsilon}_i^2}{(1-h_{ii})^2} = \left(\sum_{k=1}^n \hat{\epsilon}_k^2 \right) - \frac{\hat{\epsilon}_i^2}{1-h_{ii}}$$

$$(n-p-1) \hat{\sigma}_{[-i]}^2 = \sum_{k=1}^n \hat{\epsilon}_k^2 - \frac{\hat{\epsilon}_i^2}{1-h_{ii}}$$

$$\hat{\sigma}^2 = \frac{1}{n-p} \sum \hat{\epsilon}_i^2 \Leftrightarrow \sum \hat{\epsilon}_i^2 = (n-p) \hat{\sigma}^2$$

$$(n-p-1) \hat{\sigma}_{[-i]}^2 = (n-p) \hat{\sigma}^2 - \frac{\hat{\epsilon}_i^2}{1-h_{ii}} \quad \checkmark \text{ OK}$$

3) 2) monotone relationship btwn standardized + ~~standardized~~ studentized residual

$$\text{studr}_i = \text{standr}_i \sqrt{\frac{n-p-1}{n-p-\text{standr}_i^2}}$$

$$\sum_{k=1}^n \frac{\hat{\epsilon}_k^2}{1-h_{ii}} \geq \frac{\hat{\epsilon}_i^2}{1-h_{ii}} ;$$

$$h_{ii} + \frac{\hat{\epsilon}_i^2}{\sum_{k=1}^n \hat{\epsilon}_k^2} \leq 1$$

$$\text{standr}_i = \frac{\hat{\epsilon}_i}{\sqrt{\hat{\sigma}^2 (1-h_{ii})}}$$

$$\text{studr}_i = \frac{\hat{\epsilon}_{[-i]}}{\sqrt{\hat{\sigma}_{[-i]}^2 / (1-h_{ii})}}$$

$$= \frac{y_i - x_i^T \hat{\beta}_{[-i]}}{\sqrt{\hat{\sigma}_{[-i]}^2 / (1-h_{ii})}}$$

$$\text{from 1): } \hat{\sigma}_{[-i]}^2 = \frac{(n-p)(\hat{\sigma}^2) - \frac{\hat{\epsilon}_i^2}{1-h_{ii}}}{n-p-1}$$

$$\hat{\sigma}_{[-i]}^2 = \frac{(n-p)(\hat{\sigma}^2) - \underbrace{(\text{standr}_i)^2 \hat{\sigma}^2}}{n-p-1}$$

$$\hat{\sigma}_{[-i]}^2 = \frac{\hat{\sigma}^2 ((n-p) - \text{standr}_i^2)}{n-p-1}$$

↓ plug into studr_i

$$\text{stud}_i = \frac{\hat{\epsilon}_{[-i]}}{\sqrt{\hat{\sigma}_{[-i]}^2 / (1-h_{ii})}}$$

$$= \frac{\hat{\epsilon}_{[-i]}}{\sqrt{\frac{\hat{\sigma}^2 ((n-p) - \text{stand}_i^2)}{n+p-1}} / (1-h_{ii})}$$

$$= \frac{\hat{\epsilon}_i}{\hat{\sigma} \sqrt{1-h_{ii}}} \cdot \frac{\sqrt{n+p-1}}{\sqrt{n-p-\text{stand}_i^2}}$$

$$= \text{stand}_i \sqrt{\frac{n+p-1}{n-p-\text{stand}_i^2}} \quad \checkmark \text{ OK}$$

```
In [55]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv("Tetuan City power consumption.csv")
```

```
In [56]: df.head()
```

```
Out[56]:
```

	DateTime	Temperature	Humidity	Wind Speed	general diffuse flows	diffuse flows	Zone 1 Power Consumption	2 P Consumi
0	1/1/2017 0:00	6.559	73.8	0.083	0.051	0.119	34055.69620	16128.8
1	1/1/2017 0:10	6.414	74.5	0.083	0.070	0.085	29814.68354	19375.0
2	1/1/2017 0:20	6.313	74.5	0.080	0.062	0.100	29128.10127	19006.6
3	1/1/2017 0:30	6.121	75.0	0.083	0.091	0.096	28228.86076	18361.0
4	1/1/2017 0:40	5.921	75.7	0.081	0.048	0.085	27335.69620	17872.3

```
In [57]: features = ['Temperature', 'Humidity', 'Wind Speed',
                    'general diffuse flows', 'diffuse flows']

target = ['Zone 1 Power Consumption']
```

```
In [106... X = np.column_stack([np.ones(len(df)), df[features]])
y = df[target].values
```

```
In [107... X_train, y_train = X[:1008], y[:1008]
X_test, y_test = X[1008:2016], y[1008:2016]
```

Initial OLS (G1)

```
In [108... V = np.linalg.inv(X_train.T @ X_train)          # (X'X)^{-1}
beta_hat = V @ X_train.T @ y_train                  # OLS estimate
```

```
In [109... beta_hat
```



```
Out[109... array([[ 1.96636077e+04],
        [ 1.48061391e+03],
        [-6.33735285e+01],
        [ 1.20267864e+03],
        [-1.02262782e+01],
        [-2.03423688e+01]])
```

```
In [110... beta_hat[1][0]
```

```
Out[110... np.float64(1480.6139146660544)
```

We focus on the coefficient on temperature throughout this

```
In [111... temp_coefs = []                                # place to store temp coefficients
#temp_coefs.append(beta_hat[1][0])                # beta_0 is the intercept, beta_1 is

for i in range(len(X_test)):                      # iterate thru week 2
    x_new = X_test[i]
    y_new = y_test[i]

# (G2)
Vx = V @ x_new                                  # V x
denom = 1 + x_new @ Vx                          # 1 + x.T (V x)
V = V - np.outer(Vx, Vx) / denom

# (G3)
gamma = V @ x_new
e_hat = y_new - x_new @ beta_hat

# (G4)
beta_hat = beta_hat + gamma * e_hat

temp_coefs.append(beta_hat[1][0])
```

```
/var/folders/1r/_gd8nm2931d2rl2rndmd2_0m0000gn/T/ipykernel_39209/2935799840.
py:16: RuntimeWarning: overflow encountered in matmul
    e_hat = y_new - x_new @ beta_hat
/var/folders/1r/_gd8nm2931d2rl2rndmd2_0m0000gn/T/ipykernel_39209/2935799840.
py:19: RuntimeWarning: invalid value encountered in add
    beta_hat = beta_hat + gamma * e_hat
```

```
In [112... df2 = df[1008:2016]
```

```
In [113... len(df2)
```

```
Out[113... 1008
```

```
In [114... len(temp_coefs)
```

Out [114... 1008

In [116... df2['beta_temp'] = temp_coefs

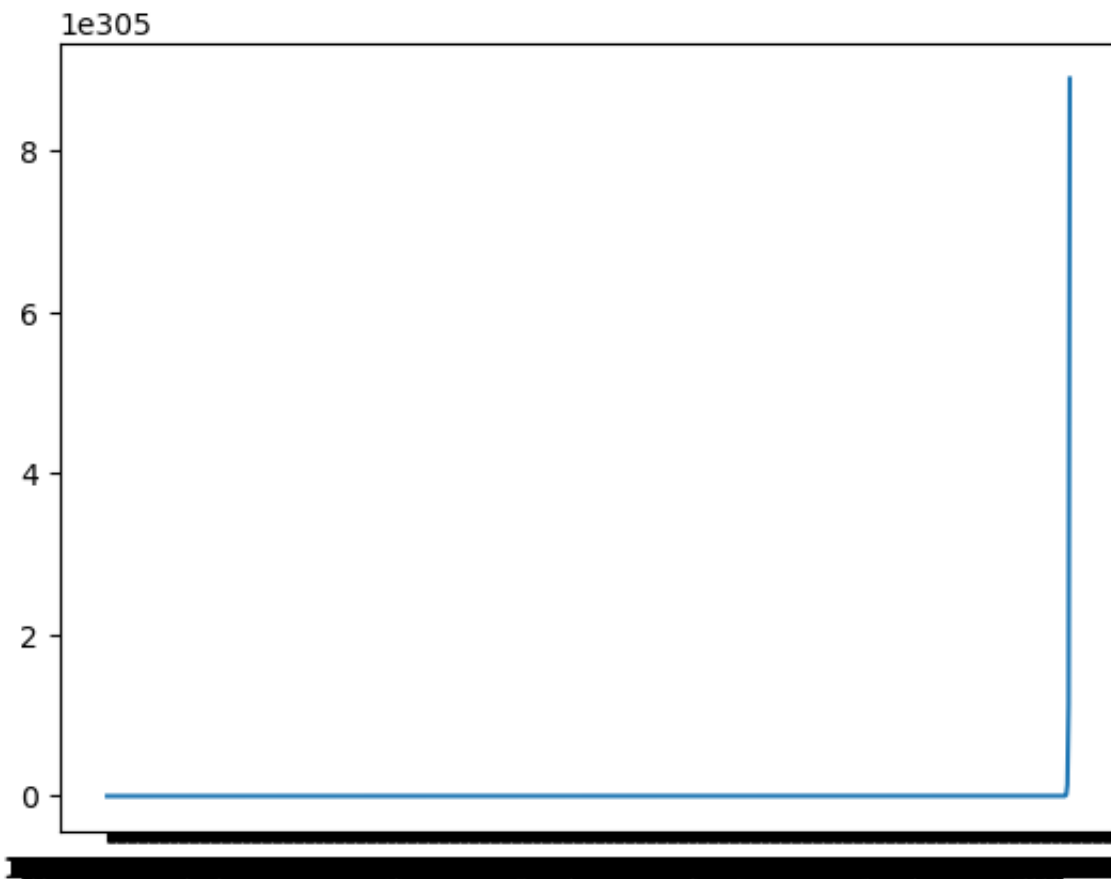
In [119... df2.sample(20)

Out [119...

	DateTime	Temperature	Humidity	Wind Speed	general diffuse flows	diffuse flows	Zone 1 Power Consumption	Cor
1063	1/8/2017 9:10	13.57	65.68	0.096	66.060	39.610	23580.75949	13
1797	1/13/2017 11:30	15.79	80.20	0.074	184.800	184.900	33952.40506	225
1231	1/9/2017 13:10	16.57	62.62	0.075	441.500	172.400	34535.69620	21
1517	1/11/2017 12:50	18.14	51.95	0.082	375.800	117.600	33885.56962	20
1922	1/14/2017 8:20	12.53	76.00	0.087	1.588	1.621	24334.17722	14
1776	1/13/2017 8:00	12.82	83.70	0.075	0.070	0.148	24540.75949	15
1835	1/13/2017 17:50	15.82	74.80	0.071	24.780	25.170	42094.17722	25
1618	1/12/2017 5:40	10.57	68.23	0.076	0.051	0.145	22936.70886	144
1355	1/10/2017 9:50	13.22	77.30	0.079	191.100	54.900	30227.84810	18
1243	1/9/2017 15:10	16.34	66.79	0.075	325.900	249.900	32324.05063	21
1865	1/13/2017 22:50	13.07	86.30	0.084	0.062	0.152	35744.81013	220
1509	1/11/2017 11:30	13.88	57.36	0.085	384.900	82.800	33290.12658	20
1062	1/8/2017 9:00	13.20	69.08	0.093	51.310	36.600	22888.10127	13
1433	1/10/2017 22:50	10.56	86.20	0.085	0.073	0.126	35732.65823	21
1258	1/9/2017 17:40	15.59	71.90	0.074	10.410	10.730	40593.41772	25

1909	1/14/2017 6:10	14.85	63.68	0.079	0.066	0.111	25093.67089	14'
1430	1/10/2017 22:20	11.13	84.60	0.084	0.026	0.145	37962.53165	23
1947	1/14/2017 12:30	15.13	66.29	0.084	146.400	146.100	34341.26582	20
1273	1/9/2017 20:10	15.30	75.90	0.074	0.022	0.130	42592.40506	26
1454	1/11/2017 2:20	10.02	83.10	0.088	0.070	0.148	23143.29114	13

```
In [117... plt.plot(df2['DateTime'], df2['beta_temp'])
plt.show()
```



```
In [122... fig, ax = plt.subplots(figsize=(11, 4.5))

steps = np.arange(len(temp_coefs))
ax.plot(steps, temp_coefs, lw=1.2,)
ax.axvline(0, color="gray", lw=0.8, ls="--")
ax.axhline(temp_coefs[0], lw=0.8, ls=":", label=f"Initial: {temp_coefs[0]:.4")
```



```
ax.axhline(temp_coefs[-1], lw=0.8, ls=":", label=f"Final: {temp_coefs[-1]}")

ax.set_xlabel("Week 2", fontsize=11)
ax.set_ylabel("Temperature coefficient", fontsize=11)
ax.set_title("Gauss Updating ", fontsize=12)
ax.legend(fontsize=10)
ax.grid(True, alpha=0.35)
fig.tight_layout()
plt.show()
```

